College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 1 | **Year:** 2012

Subject: 353152 Electronic Circuits I | **Section:** 5-6

Date: 5 October 2012 | **Time:** 10.00-12.00

Name:______________________________ ID:______________________ Field of Study:_________________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books are NOT allowed.
4. Dictionaries and a calculator are allowed.
5. No any electronic communication devices allow in the exam room.
6. The examination has 11 pages (including this page), 2 parts (55 questions) and a total score of 150 points.
7. Write all your answers on this exam sheet.

<u>Part 1</u> (100 points) Multiple choices

Instruction: Mark the correct answer for the following questions in **the provided answer sheet**.

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1. Which of following transistor operating mode represents the short circuit?

   (A) cut-off (B) saturation (C) active (D) open

2. Which class of power amplifiers that conducts less than 180° of input.

   (A) class A (B) class B (C) class AB (D) class C

3. Which class of power amplifiers having Q-point at the cutoff point.

   (A) class A (B) class B (C) class AB (D) class C

4. Which class of power amplifiers that conducts through 360° of input.

   (A) class A (B) class B (C) class AB (D) class C

5. Which of following is true?

   (A) EB Junction is forward biased when transistor is in active mode.

   (B) EB Junction is forward biased when transistor is in cutoff mode.

   (C) CB Junction is reverse biased when transistor is in saturation mode.

   (D) CB Junction is forward biased when transistor is in cutoff mode.

6. How can the class A amplifier efficiency be improved from 25%?

   (A) increase input current. (B) reduce input current.

   (C) add transformer at input side. (D) add transformer at output side.

7. What is the benefit of using Darlington pair and feedback pair transistors for push-pull operation?

   (A) reduce output swing (B) increase output power

   (C) increase input current (D) reduce distortion

8. Which type of following power amplifiers having the highest efficiency?

   (A) class A (B) class B (C) class AB (D) class C

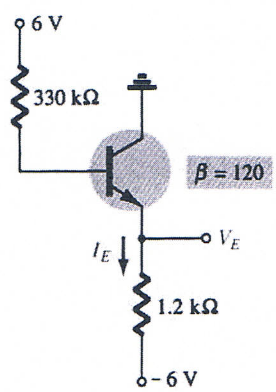


Figure 1.

9. From Figure 1, find the current I_B.

   (A) 23.78 μA (B) 23.78 mA (C) 34.11 μA (D) 34.11 mA

10. From Figure 1, find the current I_E.

    (A) 2.87 μA (B) 2.87 mA (C) 4.12 μA (D) 4.12 mA

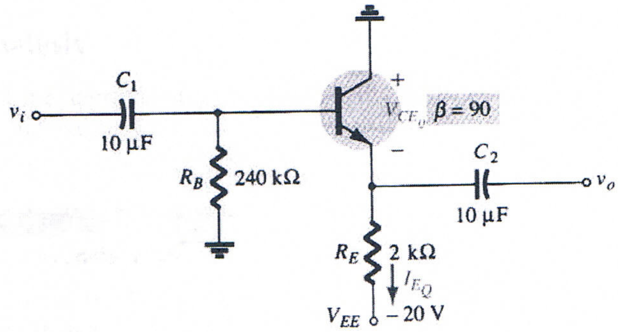


Figure 2.

11. From Figure 2, find the current I_B.

(A) 80.16 μA (B) 80.16 mA (C) 45.73 μA (D) 45.73 mA

12. From Figure 2, which of following equations is for finding V_{CE}?

(A) $V_{CE} = V_{EE} - 0.7 - I_E R_E$ (B) $V_{CE} = V_{EE} - I_E R_E$

(C) $V_{CE} = V_{EE} - I_E R_E - I_B R_B$ (D) $V_{CE} = V_{EE} - I_E R_E - 0.7 - I_B R_B$

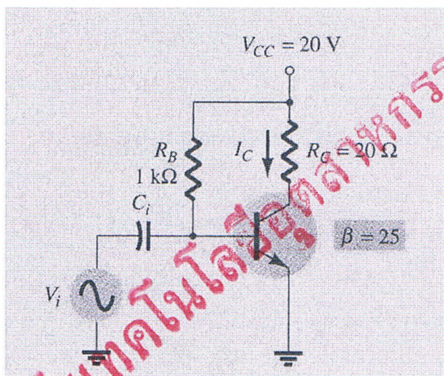


Figure 3.

13. From Figure 3, what is value of P_0(AC) if I_c = 250 mA peak?

(A) 1.25 mW (B) 1.25 W (C) 0.625 mW (D) 0.625 W

14. What is the cause of crossover distortion in push-pull circuit?

(A) two transistors do not turn on/off simultaneously.

(B) heat at transistor.

(C) two transistors are not the same type (PNP and NPN).

(D) input current is the sine wave.

15. What is the value of I_{DC} of class B amplifier if peak output current I_L(p) is 1 A.

(A) 0.318 A (B) 0.636 A (C) 0.707 A (D) 1.414 A

16. What is the value of P_0(AC) of class B amplifier if V_L(p-p) = 10 V and R_L = 2 ohms.

(A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W

17. What is the purpose of using push-pull circuit for class B amplifier?

(A) to obtain full cycle (B) to reduce the distortion

(C) to increase speed of signal (D) to reduce the cost of amplifier

18. For the most dc configurations, following law is used.

(A) KCL (B) KVL (C) Ohm (D) Lenz

19. What is the equation when the transistor is operating in saturation mode?

(A) $I_C = \dfrac{V_{CC}}{R_C}$ (B) $V_{CE} = V_{CC}$

(C) $I_B = \dfrac{V_{CC} - V_{BE}}{R_B}$ (D) $V_{CE} = V_{EE} - I_E R_E$

20. What is the equation when the transistor is operating in cut-off mode?

(A) $I_C = \dfrac{V_{CC}}{R_C}$ (B) $V_{CE} = V_{CC}$

(C) $I_B = \dfrac{V_{CC} - V_{BE}}{R_B}$ (D) $V_{CE} = V_{EE} - I_E R_E$

21. What is efficiency of class A amplifier?

(A) $\%\eta = \dfrac{P_o(dc)}{P_i(dc)} \times 100\%$ (B) $\%\eta = \dfrac{P_o(ac)}{P_i(dc)} \times 100\%$

(C) $\%\eta = \dfrac{P_o(dc)}{P_i(ac)} \times 100\%$ (D) $\%\eta = \dfrac{P_o(ac)}{P_i(ac)} \times 100\%$

22. For the most dc configuration, the analysis will start by_______.

(A) base current (B) emitter current

(C) collector current (D) emitter-collector voltage

23. Which of following circuits, emitter is connected directly to the ground.

(A) fixed-bias (B) emitter-bias

(C) emitter-follower (D) collector feedback

24. Which of the following is <u>NOT</u> the application of a transistor?

(A) Amplification of signal (B) Switching and control

(C) Computer logic circuitry (D) Fluorescents

25. In an emitter-bias configuration, the _____ the resistance R_E, the _____ the stability factor, and the _____ stable is the system.

(A) smaller, lower, less (B) larger, more, more

(C) smaller, more, more (D) larger, lower, more

26. In a transistor-switching network, the level of the resistance between the collector and emitter is _____ at the saturation and is _____at the cutoff.

(A) low, low (B) low, high (C) high, high (D) high, low

27. Calculate the delay time in a transistor switching network if t_{on} is 56 ns, t_f = 14 ns, and t_r = 20 ns.

(A) 70 ns (B) 42 ns (C) 36 ns (D) 34 ns

28. The _____the stability factor, the _____sensitive the network is to variations in that parameter.

(A) higher, more
(B) higher, less
(C) lower, more
(D) None of the above

29. In a transistor-switching network, the operating point switches from _____ to _____ regions along the load line.

(A) cutoff, active
(B) cutoff, saturation
(C) active, saturation
(D) None of the above

30. _____is the primary difference between the exact and approximate techniques used in the analysis of a voltage divider circuit.

(A) Thevenin voltage E_{Th}
(B) Thevenin resistance R_{Th}
(C) Base voltage V_B
(D) R_C

31. _____is the least stabilized circuit.

(A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback

32. _____ is less dependent on the transistor beta.

(A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback

33. The Thevenin equivalent network is used in the analysis of the _____ circuit.

(A) Fixed bias
(B) Emitter-stabilized bias
(C) Voltage divider
(D) Voltage feedback

34. The emitter resistor in an emitter-stabilized bias circuit appears to be _____ in the base circuit.

(A) larger
(B) smaller
(C) the same
(D) none of the above

35. Changes in temperature will affect the level of _____.

(A) current gain β
(B) leakage current I_{CEO}
(C) both current gain β and leakage current I_{CEO}
(D) None of the above

36. In any amplifier employing a transistor, the collector current I_C is sensitive to _____.

(A) β
(B) V_{BE}
(C) I_{CO}
(D) All of the above

37. Class D operation can achieve power efficiency of over _____.

(A) 90%
(B) 78.5%
(C) 50%
(D) 25%

38. In order to isolate the ac levels from the dc analysis in a transistor network, the capacitor can be replaced by ________.

(A) an open circuit equivalent
(B) a short circuit equivalent
(C) a voltage source
(D) a current source

39. Calculate V_{CEQ} from Figure 4.

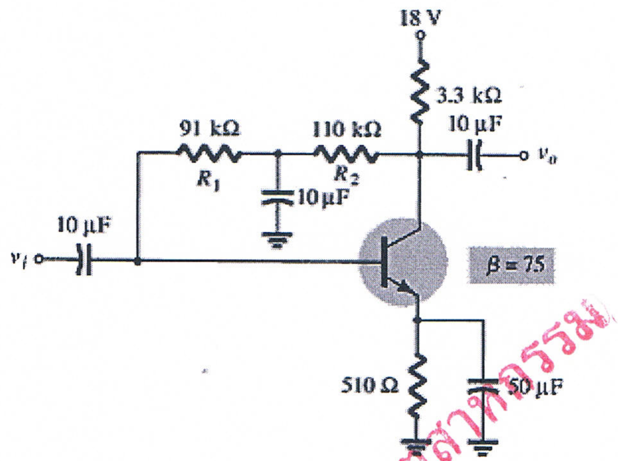


Figure 4.

(A) 8.78 V
(B) 0 V
(C) 7.86 V
(D) 18 V

40. As the temperature increases, β _____, V_{BE} _____, and I_{CO} _____ in value for every $10^{\circ}C$.

(A) increases, decreases, doubles
(B) decreases, increases, remains the same
(C) decreases, increases, doubles
(D) increases, increases, triples

41. Which type of amplifier uses pulse (digital) signals in its operation?

(A) Class A
(B) Class B or AB
(C) Class C
(D) Class D

42. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?

(A) $S(I_{CO})$
(B) $S(V_{BE})$
(C) $S(\beta)$
(D) Undefined

43. In a fixed-bias circuit, which one of the stability factors overrides the other factors?

(A) $S(I_{CO})$
(B) $S(V_{BE})$
(C) $S(\beta)$
(D) Undefined

44. Which of the following voltages must have a negative level (value) in any NPN bias circuit?

(A) V_{BE}
(B) V_{CE}
(C) V_{BC}
(D) None of the above

45. For the typical transistor amplifier in the active region, V_{CE} is usually about _____ % to _____ % of V_{CC}.

(A) 0, 100 (B) 25, 75 (C) 45, 55 (D) 50, 50

46. In a collector feedback bias circuit, the current through the collector resistor is _____ and the collector current is _____.

(A) I_C , I_C
(B) $I_B + I_C$, I_C
(C) I_B , I_C
(D) None of the above

47. For a transistor operating in the saturation region, the collector current I_C is at its _____ and the collector-emitter voltage V_{CE} is to the _____.

(A) minimum, left of the V_{CEsat} line
(B) minimum, right of the V_{CEsat} line
(C) maximum, left of the V_{CEsat} line
(D) maximum, right of the V_{CEsat} line

48. How many transistors must be used in a class B power amplifier to obtain the output for the full cycle of the signal?

(A) 0 (B) 1 (C) 2 (D) 3

49. A heat sink provides _____ thermal resistance between case and air.

(A) a high
(B) a low
(C) the same
(D) None of the above

50. A _____ power amplifier is limited to use at one fixed frequency.

(A) class A
(B) class B or AB
(C) class C
(D) class D

Part 2 (25 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

1. From Figure 5, calculate the input power $P_i(dc)$ and output power $P_o(ac)$ of the circuit. The input signal results in a base current of 5 mA_{RMS}, where $\beta = 50$ and $V_{cc} = 20\ V$

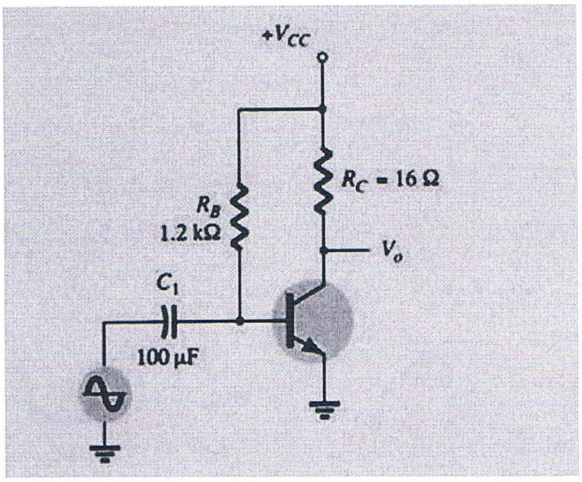


Figure 5.

2. From Figure 6, determine the level of V_{CE} and I_B and for the network with $\beta = 100$.

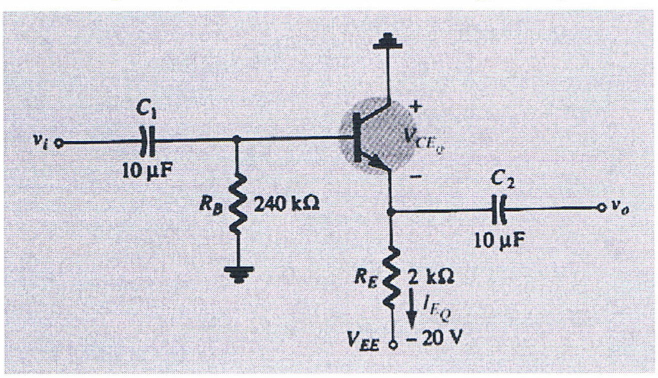


Figure 6.

3. From Figure 7, determine I_B, V_B, and V_E with $\beta = 100$

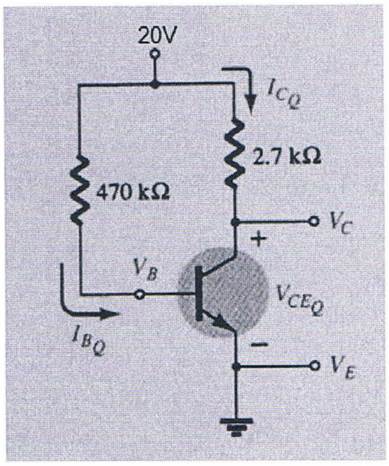


Figure 7.

4. Use this table below to determine the change in I_C from 25°C to 100°C for R_B / R_E = 70 due to the $S(I_{CO})$ stability factor. Assume an emitter-bias configuration where $S(I_{CO}) = \frac{(\beta+1)\left(1+\frac{R_B}{R_E}\right)}{(\beta+1)+\frac{R_B}{R_E}}$.

T (°C)	I_{CO} (nA)	β	V_{BE}(V)
−65	0.2×10^{-3}	20	0.85
25	0.1	50	0.65
100	20	80	0.48
175	3.3×10^{3}	120	0.3

5. Calculate R_{th} , E_{th} and V_B from Figure 8.

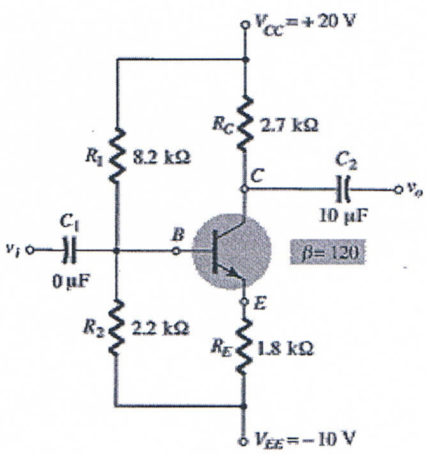
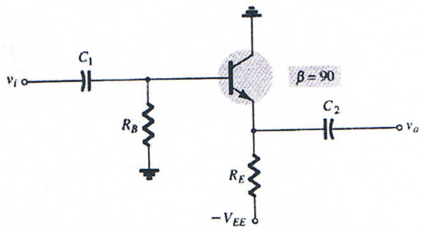


Figure 8.

Handout

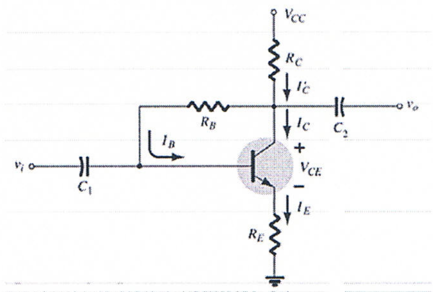
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

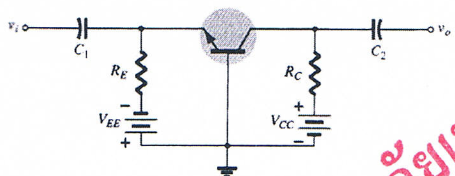
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_0(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\%h = 50\left(\frac{V_{CE(\max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(\min)}}\right)^2$$

$$\frac{R_L'}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2}\right)^2 = a^2$$

Class B amplifier

$$P_0(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC}\left(\frac{2}{\pi} I_L(p)\right)$$

Answer Sheet

เลขที่นั่งสอบ

Name:______________________________ ID:____________________

Subject: 342152 Electronic Circuits I

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